EC 1745 Midterm: The Basics, Step by Step

This guide assumes you are starting from zero. Each section explains one topic in plain English, then shows exactly what to type into Excel, then works a real question from the 2024 or 2025 practice midterm. Your notes sheet is the short version of this guide.

How to use it. Read one section, then do the matching practice question from a blank workbook without looking. Check your answer against the answer key. If it is wrong, find the step where you went off.

Section	Topic	Practice with
0	Excel in 10 minutes	Everything
1	Income statement	2025 Part 2; mock Part 1
2	Balance sheet	Mock Part 1
3	Cash flow statement	Mock Part 1
4	Ratios, DuPont, growth, firm value	2025 Part 2; 2024 Part 3
5	Time value of money, loans, rates	2025 Part 3; mock Part 2
6	Bonds (with and without dates)	2024 Parts 4–5; 2025 Part 4; mock Parts 3–4
7	Duration	Same as bonds
8	Exam-day checklist	

0. Excel in 10 minutes

Cells and formulas

- Every box is a **cell**, named by its column letter and row number. B3 is column B, row 3.
- Typing **1000** stores a number. Typing **=B2-B3** stores a **formula**: Excel calculates it and shows the result. Every formula starts with an equals sign.
- Math symbols: + add, – subtract, * multiply, / divide, ^ power. =1.1^3 means 1.1 cubed.
- Excel follows normal math order, so use parentheses: **=100/(1+0.1)^3** gives 75.13.

Why you point at cells instead of typing numbers

If B2 holds revenue and B3 holds cost, write **=B2-B3**, not =1000-600. If you mistype an input, you fix it once and every answer updates. The grader can also see your method, which earns partial credit.

Copying formulas, and the $ sign

- Select a cell and press Ctrl+C, then paste somewhere else. Excel shifts the cell references: =B2-B3 copied one column right becomes =C2-C3. That is how you fill in a second year quickly. Ctrl+R fills right and Ctrl+D fills down.
- To stop a reference from shifting, lock it with $: **=B5*(1+$B$1)**. Click inside the reference and press **F4** (Mac: Cmd+T) to add the dollar signs. Use this for a discount rate or tax rate that every row uses.

Percentages, dates and functions

- Store rates as decimals: 5% is 0.05. You can also type 5% and Excel stores 0.05. Click the % button (Home tab) to display a cell as a percent. 1 basis point (bp) = 0.01% = 0.0001, so 100 bp = 0.01.
- Dates: type **=DATE(2025,10,7)** for October 7, 2025. If a cell shows a number like 45937, it is a date stored as a number. Format it as a date.

- A **function** is a built-in formula: **=SUM(B2:B10)** adds B2 through B10. Type =PV(and Excel shows a hint listing the inputs in order. Inputs in [square brackets] are optional.

Excel's minus-sign rule for finance functions

PV, FV, PMT, RATE and NPER treat money you pay and money you receive as opposite signs. **=PV(0.1,10,30)** returns −184.34, the amount you would pay today to receive 30 a year. To get a positive answer, put a minus sign on the payment: **=PV(0.1,10,-30)** = 184.34. If RATE or PMT returns #NUM!, your signs are probably the same. Make one of them negative.

How to lay out each exam tab

Put inputs at the top, each with a label in column A and its value in column B. Put answers below, each next to its question label. Example:

Cell	Column A	Column B
row 1	Discount rate	0.10 (typed)
row 2	Payment per year	30 (typed)
row 3	Years	10 (typed)
row 5	3.B.a PV of contract 1	**=PV(B1,B3,-B2)** → 184.34

1. The income statement

The income statement shows how a company did over one year. It starts with sales at the top (the **top line**). Each kind of cost is subtracted in order. What is left at the bottom is net income, the profit that belongs to shareholders (the **bottom line**).

What each line means

Line	What it is	How to get it
Revenue (sales)	Money from customers. The top line.	Given
− Cost of goods sold (COGS)	Direct cost of making or buying what was sold	Given
= Gross profit	What is left after direct costs	Revenue − COGS
− SG&A	Selling, general and administrative: salaries, rent, marketing	Given
= EBITDA	Earnings before interest, taxes, depreciation and amortization	Gross profit − SG&A
− Depreciation & amortization	A long-lived asset's cost spread over its life. No cash leaves.	Given
= EBIT	Operating profit: profit from the business before financing and taxes	EBITDA − depreciation
− Interest expense	The cost of the company's debt	Given
= Pretax income (EBT)	Profit before taxes	EBIT − interest
− Taxes	Tax rate × pretax income	Rate × EBT
= Net income	Profit for shareholders. The bottom line.	EBT − taxes
− Dividends	Cash paid out to shareholders	Given

= Retained earnings	Profit kept in the company. It adds to equity.	NI – dividends

Build it in Excel (2025 exam, Ferrario)

Row	Column A	Column B (2024)	Result
1	Year	2024	
2	Revenue	1000	1,000
3	Cost of goods sold	600	600
4	Gross profit	**=B2-B3**	400
5	SG&A	100	100
6	EBITDA	**=B4-B5**	300
7	Depreciation	60	60
8	EBIT	**=B6-B7**	240
9	Interest expense	25	25
10	Pretax income	**=B8-B9**	215
11	Tax rate	0.21	21%
12	Taxes	**=B11*B10**	45.15
13	Net income	**=B10-B12**	169.85
14	Dividends	60	60
15	Retained earnings	**=B13-B14**	109.85

A second year goes in column C. Type its inputs, then copy the formulas from column B into column C.

Mistakes to avoid

- Taxes are charged on **pretax income (EBT)**, not on EBIT or revenue.
- Depreciation is subtracted **after** EBITDA and **before** EBIT.
- "Net income" and "retained earnings" are different. Retained earnings = net income – dividends.

2. The balance sheet

The balance sheet is a snapshot on one date (for example, December 31). The left side lists what the company **owns** (assets). The right side shows who paid for those assets: lenders and suppliers (**liabilities**) and the owners (**equity**). The two sides are always equal:

Total assets = total liabilities + shareholders' equity

What it looks like

ASSETS (what the firm owns)	LIABILITIES & EQUITY (who paid for it)
Cash	Accounts payable (AP): bills owed to suppliers
+ Accounts receivable (AR): customers owe you	+ Short-term debt: loans due within a year
+ Inventory: goods not yet sold	**= Total current liabilities**
= Total current assets	+ Long-term debt
+ Gross PP&E (property, plant & equipment)	**= Total liabilities**

– Accumulated depreciation	+ Shareholders' equity
= Net PP&E	
+ Other assets	
= TOTAL ASSETS	**= TOTAL LIABILITIES + EQUITY**

Build it in Excel (2024 exam, Bergstresser, 2024 column)

In Excel, stack the two sides in one column so the formulas are simple:

Row	Column A	Column B	Result
2	Cash	20	20
3	Accounts receivable	83	83
4	Inventory	60	60
5	Total current assets	**=SUM(B2:B4)**	163
6	Gross PP&E	2028	2,028
7	Accumulated depreciation	1033	1,033
8	Net PP&E	**=B6-B7**	995
9	Other assets	900	900
10	TOTAL ASSETS	**=B5+B8+B9**	2,058
12	Accounts payable	53	53
13	Short-term debt	82	82
14	Total current liabilities	**=B12+B13**	135
15	Long-term liabilities	897	897
16	Total liabilities	**=B14+B15**	1,032
17	Shareholders' equity	1026	1,026
18	TOTAL LIABILITIES + EQUITY	**=B16+B17**	2,058 (matches row 10)

Constructing a balance sheet when something is missing

The exam may give you last year's balance sheet, this year's income statement, and most of this year's balance sheet. Then you fill in the gaps with three rules.

1. **Equity rolls forward.** Equity this year = equity last year + retained earnings this year (+ new stock issued – stock bought back). Example (mock exam): 850 + 180 = 1,030.
2. **Net PP&E rolls forward.** Net PP&E this year = net PP&E last year + CapEx – depreciation.
3. **The plug.** If one item is still unknown (usually cash), use assets = liabilities + equity. Cash = (total liabilities + equity) – (all other assets). Mock exam: 1,750 – (230 + 280 + 1,080) = 160.

If a question gives only debt and equity (as in the 2025 exam: equity 1,200, debt 500, no other liabilities), then total assets = 500 + 1,200 = 1,700.

3. The cash flow statement

Net income is not the same as cash. Depreciation is subtracted as a cost but no cash leaves. A sale on credit counts as revenue before the customer pays. Inventory costs cash before it is sold. The cash flow statement fixes these differences and splits cash into three buckets.

Δ (delta) means change: this year minus last year. ΔAR = AR 2025 − AR 2024.

The template

Line	Why	Mock exam (Pemberton)
Net income	Start from profit	240
+ Depreciation	Was subtracted, but no cash left	+120
− ΔAR	More AR = customers haven't paid yet	−30
− ΔInventory	More inventory = cash tied up in goods	−30
+ ΔAP	More AP = you haven't paid suppliers yet	+20
= Cash from operations (CFO)		**320**
− CapEx	Spending on PP&E. CapEx = ΔNet PP&E + depreciation	−200 (80 + 120)
= Cash from investing (CFI)		**−200**
+ ΔDebt + new stock − dividends	Money from or to lenders and owners	+50 − 60
= Cash from financing (CFF)		**−10**
CFO + CFI + CFF	Must equal the change in cash on the balance sheet	**110** = 160 − 50 ✓

Rule of thumb. An asset going up uses cash (subtract it). A liability going up provides cash (add it).

Free cash flow (FCF)

FCF is the cash the business produces after paying for the investment it needs. It is available to all investors (lenders and owners).

FCF = EBIT × (1 − tax rate) + depreciation − CapEx − ΔNWC, where ΔNWC = ΔAR + ΔInventory − ΔAP.

Mock exam: 360 × 0.75 + 120 − 200 − 40 = 150. In Excel, each piece is a cell and FCF is one formula referencing them.

4. Ratios, DuPont, growth and firm value

A ratio divides one number by another so you can compare companies of different sizes, or one company across years. Each ratio goes in its own labeled cell as a formula.

Ratio	Formula	What it tells you	Ferrario 2024
Gross margin	Gross profit / Revenue	Cents kept per $1 of sales after direct costs	40.0%
Profit margin	Net income / Revenue	Cents of profit per $1 of sales	16.99%
ROE	Net income / Equity	Profit per $1 the owners put in (book value)	14.15%
ROA	Net income / Total assets	Profit per $1 of assets	9.99%
Asset turnover	Revenue / Total assets	Sales per $1 of assets (efficiency)	0.588
Equity multiplier	Total assets / Equity	Assets per $1 of equity (leverage)	1.417
Current ratio	Current assets / Current liabilities	Can it pay bills due this year?	
Interest coverage	EBIT / Interest	How many times profit covers interest	9.6x

Debt-to-equity	Total debt / Equity	How much is borrowed vs. owned	0.417

DuPont: why ROE is what it is

ROE = profit margin × asset turnover × equity multiplier

(NI / Revenue) × (Revenue / Assets) × (Assets / Equity). Revenue and assets cancel out, leaving NI / Equity. Ferrario: 16.99% × 0.588 × 1.417 = 14.15%, the same as ROE. If the product does not match, something is wrong.

To interpret, say which piece moved. Bergstresser's ROE rose from 10.1% (2023) to 15.7% (2024) because profit margin rose from 1.6% to 2.6%. Turnover (about 3.0) and leverage (about 2.0) barely changed.

Growth rates

1. Retention ratio **b** = (net income − dividends) / net income. Ferrario: 109.85 / 169.85 = 0.647.
2. **Sustainable growth rate** = ROE × b = 14.15% × 0.647 = 9.15%. This is how fast the company can grow using retained earnings plus new borrowing that keeps debt/equity constant, with no new stock.
3. **Internal growth rate** = ROA × b = 9.99% × 0.647 = 6.46%. This is how fast it can grow using only retained earnings, with no outside money.

Some textbooks use ROE·b / (1 − ROE·b) instead. Check your lecture notes. If unsure, write the version you used next to the answer.

What the firm is worth

- **Market capitalization** = shares × share price. This is the market value of equity.
- **Enterprise value (EV)** = market cap + debt − excess cash. It is the value of the whole business, to lenders and owners together.
- If the question gives EV as a multiple ("EV is 8 times EBIT"), work backwards: EV = 8 × 240 = 1,920. Equity = EV − debt + cash = 1,920 − 500 + 0 = 1,420.
- **P/E** = market value of equity / net income = 1,420 / 169.85 = 8.36. Same as share price / EPS.

5. Time value of money

$100 today is worth more than $100 next year, because you could invest today's $100 and earn interest. The **discount rate r** is that interest rate. Discounting converts future money into today's money.

One cash flow

- **Present value:** PV = CF / (1 + r)^t. $100 in 3 years at 10%: **=100/(1.1)^3** = 75.13.
- **Future value:** FV = PV × (1 + r)^t. $75.13 today grows to **=75.13*1.1^3** = 100 in 3 years.

Many cash flows: always draw a timeline

t = 0 is today, t = 1 is one year from now. Write each cash flow under its year. The PV of the whole stream is the sum of the PVs of each piece.

The "dumb" way: a timeline in Excel (2025 exam, contract 1)

Row	What goes in it	Formula in column C (copy right to L)
1	Discount rate in B1	0.10 typed in B1
3	Year t (B3 = 0, C3 = 1, … L3 = 10)	type 0 to 10
4	Cash flow	30 in C4 to L4
5	Discount factor 1/(1+r)^t	**=1/(1+$B$1)^C3**

6	PV of that year's cash flow	**=C4*C5**
8	Total PV	**=SUM(B6:L6)** → 184.34

=SUMPRODUCT(B4:L4,B5:L5) does rows 6 and 8 in one step. To get the value at year 10 instead of today, use **=C4*(1+$B$1)^(10-C3)** in each column and sum, or multiply the PV by 1.1^10.

The "smart" way: Excel functions

Function	Use it for	Example
=PV(rate, nper, pmt)	Value today of equal payments	**=PV(0.1,10,-30)** = 184.34
=FV(rate, nper, pmt)	Value at the end of equal payments	**=FV(0.1,10,-30)** = 478.12
=PMT(rate, nper, pv)	Equal payment with a given PV	**=PMT(0.08,15,-85.59)** = 10
=RATE(nper, pmt, pv)	Discount rate implied by a deal	**=RATE(15,10,-90)** = 7.19%
=NPER(rate, pmt, pv)	How many payments	**=NPER(0.08,10,-85.59)** = 15
=NPV(rate, CF1:CFn)	PV of uneven flows starting at t = 1	Add the t = 0 cash flow yourself
=IRR(CF0:CFn)	Rate that makes NPV = 0	CF0 must be negative

The 2025 contracts: contract 1 pays 30 for 10 years, PV 184.34. Contract 2 pays 35 for 8 years, PV **=PV(0.1,8,-35)** = 186.72. Contract 2 pays less in total (280 vs. 300) but is worth more, because the money arrives sooner. Value at year 10: **=FV(0.1,10,-30)** = 478.12 and **=FV(0.1,8,-35)*1.1^2** = 484.31. Contract 2's money sits and grows for the last 2 years.

Shortcut formulas

- **Annuity** (same payment C each year for N years, first payment in 1 year): PV = C / r × (1 − 1 / (1 + r)^N)
- **Growing annuity** (first payment C_1, growing at g): PV = C_1 / (r − g) × (1 − ((1 + g) / (1 + r))^N)
- **Perpetuity** (forever): PV = C / r. Growing perpetuity: PV = C_1 / (r − g).
- **Annuity due** (first payment today): multiply the ordinary annuity PV by (1 + r). In Excel, add ,0,1 at the end: **=PV(0.08,15,-10,0,1)**.

Interest rates and loans

- An **APR** is a quoted annual rate. Divide by the number of periods to get the rate per period: 7.2% APR with monthly payments = 0.6% per month.
- The **effective annual rate** includes compounding: EAR = (1 + APR/k)^k − 1. 7.2% compounded monthly: **=(1+0.072/12)^12-1** = 7.44%, or **=EFFECT(0.072,12)**.
- **Loan payment:** $90m over 10 years at 7.2% APR, monthly: **=PMT(0.072/12,120,-90)** = 1.0543 per month.
- **Balance owed** after 36 payments = PV of the 84 payments still due: **=PV(0.072/12,84,-1.0543)** = 69.40.

Restructuring a commitment

A restructured deal is fair when its PV equals the old deal's PV at the same discount rate. So find the old PV, then solve for the unknown in the new deal with PMT (payment), RATE (rate) or NPER (length). Example: a settlement of 10 per year for 15 years at 8% is worth **=PV(0.08,15,-10)** = 85.59. If the plaintiff wants 90 today instead, the implied rate is **=RATE(15,10,-90)** = 7.19%. At Corrigan's 8% rate, paying 90 costs more than 85.59, so Corrigan prefers the annual payments.

6. Bonds

A bond is a loan that investors can buy and sell. The company (the issuer) pays a fixed **coupon** every six months and repays the **par value** (also called face value) at **maturity**. Prices are quoted per 100 of par.

Example: what a bond pays

A 10-year bond with a 5% coupon, paid semiannually, pays 5% × 100 / 2 = **2.50 every six months for 20 periods**, plus **100 at the end**.

Price and yield

- The bond's price is the PV of those cash flows. The rate that makes PV equal the price is the **yield to maturity (YTM)**. It is quoted as an APR, so use **y/2 per six-month period**.
- Price and yield move in **opposite directions**. When market yields rise, existing bonds' fixed coupons look worse, so their price falls.
- Coupon rate > yield → price above 100 (premium). Coupon = yield → price = 100 (par). Coupon < yield → price below 100 (discount). A bond **issued at par** has coupon = yield.
- **Corporate yield = Treasury yield + spread.** The spread pays investors for default risk. 100 bp = 1% = 0.01.
- **Effective annual yield** = (1 + y/2)^2 − 1. A 6% APR yield → 1.03^2 − 1 = 6.09%.

Bonds without dates: the "dumb" way (2024 exam, Dioguardi)

Issued 3 years ago with 20 years to maturity at a price of 94.221 and yield 6%. Semiannual.

Question	Formula	Answer
Coupon per half-year	**=PMT(0.06/2, 40, -94.221, 100)**	2.75
Annual coupon rate	2 × 2.75 / 100	5.50%
Periodic yield; EAY	0.06/2; (1.03)^2 − 1	3%; 6.09%
Price now at 7% (17 years left = 34 periods)	**=PV(0.07/2, 34, -2.75, -100)**	85.224
Price if yield jumps to 8%	**=PV(0.08/2, 34, -2.75, -100)**	76.986
Approximate duration	−((76.986 − 85.224) / 85.224) / 0.01	9.67

To solve for a yield instead: **=2*RATE(periods, coupon, -price, 100)**.

Bonds with dates: the "smart" way

Real bonds are bought between coupon dates. The **settlement date** is the day you buy. The seller has earned part of the next coupon already, so you pay them that **accrued interest** on top of the quoted price.

- **Clean price** = the price dealers quote. **PRICE()** returns the clean price.
- **Dirty price** = clean price + accrued interest. This is what you actually pay.
- **Accrued interest** = (coupon × 100 / 2) × (days since last coupon / days in the coupon period).
- **Day-count basis.** Basis 0 = 30/360: every month counts as 30 days, so a half-year is 180 days (corporate bonds). Basis 1 = actual/actual: real calendar days (Treasuries). The question tells you which.

Step by step (2025 exam, Taglialatela)

Settlement 10/7/2025. Treasury: matures 8/15/2035, 4% coupon, clean price 98, basis 1. Corporate: matures 9/15/2035, 5% coupon, basis 0, spread 100 bp. In Excel, put each input in its own cell. Formulas are written out here with the numbers so you can see them.

Step	Formula	Answer
1. Treasury yield	**=YIELD(DATE(2025,10,7), DATE(2035,8,15), 0.04, 98, 100, 2, 1)**	4.250%
2. Corporate yield	Treasury yield + 0.01	5.250%
3. Clean price	**=PRICE(DATE(2025,10,7), DATE(2035,9,15), 0.05, 0.0525, 100, 2, 0)**	98.080
4. Days since last coupon	**=COUPDAYBS(settle, maturity, 2, 0)**	22
5. Days in the period	**=COUPDAYS(settle, maturity, 2, 0)**	180
6. Accrued interest	2.50 × 22 / 180	0.3056
7. Dirty price	98.080 + 0.3056	98.386
8. Macaulay duration	**=DURATION(settle, maturity, 0.05, 0.0525, 2, 0)**	7.905
9. Modified duration	**=MDURATION(settle, maturity, 0.05, 0.0525, 2, 0)**	7.703
10. Price change if yield +100 bp	−7.703 × 0.01	≈ −7.70%

Use the full-precision Treasury yield from step 1 in step 3 (point at the cell), not the rounded 5.25%. Other helpers: **COUPPCD** and **COUPNCD** give the previous and next coupon dates (format the cell as a date), and **COUPNUM** gives the number of coupons left.

Bond issuance and exchanges

- To raise $X by selling bonds at price P (per 100), you must issue face value = X / (P / 100).
- Market value of outstanding debt = face outstanding × P / 100.
- **Exchange offer:** the new bond must be worth the same as the old bond at the market yield. Find the old bond's price, then solve for the new coupon: **=2*PMT(y/2, 2*years, -old price, 100)/100**.

7. Duration

What it is

Duration is the **average time until you receive a bond's cash, weighted by how much each payment is worth today**. A zero-coupon bond pays everything at maturity, so its duration equals its maturity. A coupon bond pays some cash earlier, so its duration is shorter than its maturity.

Why it matters

Duration measures **interest-rate risk**: how much the bond's price moves when yields change. A bond with modified duration 7.7 loses about 7.7% of its value if yields rise by 1 percentage point. Investors use duration to compare how risky bonds are, and to match the duration of their assets to the duration of their liabilities (immunization). Banks, pension funds and insurers do this.

- **Macaulay duration** D = Σ (t × PV of cash flow at t) / price, with t in years.
- **Modified duration** MD = D / (1 + y/2) for semiannual bonds (D / (1 + y) for annual).
- **% change in price ≈ −MD × change in yield.**

Computing it by hand: a simple example

A 3-year bond with a 10% **annual** coupon and a 10% yield. Its cash flows are 10, 10 and 110.

t (years)	Cash flow	PV = CF / 1.1^t	t × PV

1	10	9.091	9.091
2	10	8.264	16.529
3	110	82.645	247.934
Sum		**100.000** = price	**273.554**

1. Macaulay D = 273.554 / 100 = **2.736 years**.
2. Modified duration = 2.736 / 1.10 = **2.487**.
3. If the yield rises from 10% to 11%, the price changes by about −2.487 × 0.01 = **−2.49%**.
4. Actual price at 11%: **=PV(0.11,3,-10,-100)** = 97.556, a change of −2.44%. The real drop is a little smaller than the estimate. That gap is called **convexity**.

In Excel, the dumb way and the smart way

- **Dumb:** one row per payment. Columns for period k, time in years (=k/2 for semiannual), cash flow, PV **=CF/(1+y/2)^k**, and t × PV. D = SUM(t × PV) / SUM(PV). If you are between coupon dates, the first period is a fraction: periods to coupon k = k − 1 + COUPDAYSNC / COUPDAYS.
- **Smart:** **=DURATION(settle, maturity, coupon, yield, 2, basis)** and **=MDURATION(...)**.
- **No dates given?** Make up two dates exactly that many years apart: **=DURATION(DATE(2000,1,1), DATE(2017,1,1), 0.055, 0.07, 2)** for a 17-year bond.
- **From two prices:** D ≈ −(% change in price) / (change in yield). This is what 2024 Part 4.D asked for.

Rules of thumb

- Longer maturity → higher duration. Higher coupon → lower duration. Higher yield → lower duration.
- The duration estimate works best for small yield changes. The real price always ends up a bit higher than the estimate: it falls less when yields rise and rises more when yields fall.

8. Exam-day checklist

1. Save the file as LASTNAME_FIRSTNAME_EC1745MIDTERM2026.xlsx (or the name the exam asks for). Type your name in cell A1.
2. Make one tab per part. Name the tabs “Part 2”, “Part 3”, and so on.
3. Type every given number once, in a labeled input cell. Every answer is a formula that points at those cells.
4. Put each answer next to its question label (for example “2.C.a Profit margin”) and highlight it.
5. Press Ctrl+S after every part.
6. Run the checks. The DuPont product equals ROE. The balance sheet balances. CFO + CFI + CFF equals the change in cash. The dumb and smart methods give the same answer.
7. If a question is ambiguous (which growth-rate formula, which equity number), pick one, and type a short note saying which one you used.
8. Bring the notes sheet with your name on it. You hand it in at the end.